

Problem5

May 3, 2026

0.1 2.5 Machine maintenance

States: R (running), B (broken) at the start of each week.

Terminal: $J_5(R) = J_5(B) = 0$

Immediate expected net profits and transitions:

- R + Maintain: $0.6(100 - 20) + 0.4(0 - 20) = 40$, next state: R w.p. 0.6, B w.p. 0.4
- R + No maintain: $0.3(100) + 0.7(0) = 30$, next state: R w.p. 0.3, B w.p. 0.7
- B + Repair: $0.6(100 - 40) + 0.4(0 - 40) = 20$, next state: R w.p. 0.6, B w.p. 0.4
- B + Replace: $1(100 - 150) = -50$, next state: R w.p. 1

Week 4:

$J_4(R)$: Maintain $\rightarrow 40$, No maintain $\rightarrow 30 \Rightarrow J_4(R) = \max(40, 30) = \mathbf{40}$ (Maintain)

$J_4(B)$: Repair $\rightarrow 20$, Replace $\rightarrow -50 \Rightarrow J_4(B) = \max(20, -50) = \mathbf{20}$ (Repair)

Week 3:

$J_3(R)$: Maintain $\rightarrow 40 + 0.6(40) + 0.4(20) = 72$, No maintain $\rightarrow 30 + 0.3(40) + 0.7(20) = 56$
 $\Rightarrow J_3(R) = \max(72, 56) = \mathbf{72}$ (Maintain)

$J_3(B)$: Repair $\rightarrow 20 + 0.6(40) + 0.4(20) = 52$, Replace $\rightarrow -50 + 40 = -10 \Rightarrow J_3(B) = \max(52, -10) = \mathbf{52}$ (Repair)

Week 2:

$J_2(R)$: Maintain $\rightarrow 40 + 0.6(72) + 0.4(52) = 104$, No maintain $\rightarrow 30 + 0.3(72) + 0.7(52) = 88$
 $\Rightarrow J_2(R) = \max(104, 88) = \mathbf{104}$ (Maintain)

$J_2(B)$: Repair $\rightarrow 20 + 0.6(72) + 0.4(52) = 84$, Replace $\rightarrow -50 + 72 = 22 \Rightarrow J_2(B) = \max(84, 22) = \mathbf{84}$ (Repair)

Week 1 (new machine, guaranteed to run $\Rightarrow$ earns \$100, ends in state R, no maintain needed):

$$J_1 = 100 + J_2(R) = 100 + 104 = \boxed{204}$$